

PROBLEM

1.A* SEARCH ALGORITHM

Problem Statement

The A* (A-Star) Search Algorithm is one of the most efficient informed search algorithms used in Artificial Intelligence for finding the shortest and optimal path between a start node and a goal node. It combines the actual path cost from the start node $g(n)$ with the estimated cost to reach the goal $h(n)$ to compute the evaluation function $f(n) = g(n) + h(n)$. By considering both the path cost and heuristic estimate, A* efficiently explores the search space while avoiding unnecessary paths.

In this problem, a weighted graph with heuristic values is provided. The objective is to perform a dry run of the A* Search Algorithm to find the optimal path from the start node A to the goal node G. During each iteration, the algorithm should display the current node, Open List, Closed List, $g(n)$, $h(n)$, and $f(n) = g(n) + h(n)$ values. After completing the search, the program should identify the optimal path and calculate the total path cost, demonstrating how the A* algorithm uses heuristic information to efficiently determine the shortest path.

2.MINIMAX WITH ALPHA-BETA PRUNING

Problem Statement

The Minimax Algorithm is a fundamental decision-making algorithm used in Artificial Intelligence for two-player, zero-sum games where one player tries to maximize the score while the other tries to minimize it. As the size of the game tree increases, evaluating every possible node becomes time-consuming and computationally expensive. Alpha-Beta Pruning is an optimization technique that improves the efficiency of the Minimax algorithm by pruning branches that cannot affect the final decision, thereby reducing the number of nodes evaluated.

In this problem, a game tree is provided with a MAX node as the root and MIN nodes as its children. The leaf nodes contain utility values representing the outcome of the game. The objective is to perform a dry run of the Minimax Algorithm with Alpha-Beta Pruning by evaluating the tree from left to right. During the execution, the program should display the Alpha (α) and Beta (β) values at each node, determine the selected minimax value, and identify any branches or nodes that are pruned using the Alpha-Beta pruning condition.